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1 Monitoring Foundry Agents — Per-User Token Tracking via APIM AI Gateway

Track **who** uses **which agent**, how many **tokens** are consumed, and the **estimated cost** — all with per-caller attribution that Foundry's built-in tracing does not natively provide.

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1.2 Overview

1.2.1 What Foundry Already Provides (Built-In)

Microsoft Foundry has **built-in observability** for agents. Once you connect an Application Insights resource to your Foundry project, server-side traces are captured automatically for:

- **Prompt Agents** — tracing is **generally available (GA)**
- **Hosted Agents, Workflow Agents, Custom Agents** — tracing is in **public preview**

Built-in tracing captures per-agent: - User inputs and agent outputs - Tool usage (tool calls, arguments, results) - **Token consumption** (input tokens, output tokens, total tokens) - Latency and duration per span - Error tracking and retries

This data is stored in Application Insights and viewable in the Foundry portal under **Agents** → **Traces** (last 90 days). For client-side visibility, you can instrument your code with the `azure-ai-projects`

SDK and `AIProjectInstrumentor`.

> **Reference**: Trace agent concepts · Setup tracing · Client-side tracing

1.2.2 What Foundry Does NOT Provide

Foundry's native tracing operates **per-agent** — it knows which agent ran, which model was called, and how many tokens were consumed. But it does **not** natively:

- Attribute usage to **individual callers** (which user, team, or application triggered the call)
- Enforce **per-user quotas** or rate limits (Agent Service has no API rate limits — only model deployment limits)
- Provide **cost allocation** per caller/team
- Add **RBAC-based access control** to specific agents (any authenticated user can call any agent in the project)

1.2.3 What APIM Adds

By routing all agent traffic through **Azure API Management (APIM)** acting as an **AI Gateway**, every request is enriched with caller metadata and — critically — **token usage is extracted server-side from the Foundry response by APIM outbound policies**, then correlated with the caller identity.

This means:

- **Per-caller attribution** — who consumed how many tokens on which agent
- **Zero client instrumentation** — callers just call APIM, no tracking SDK needed
- **Tamper-proof** — metrics are emitted by the gateway, not by the consumer
- **Universal** — any client (Python, C#, curl, Power Automate) is automatically tracked
- **Per-user quotas and rate limits** — enforced at the gateway level
- **Agent-level RBAC** — via APIM Products and Subscriptions
- **Cost estimation** per request, per user, per model

1.3 Architecture

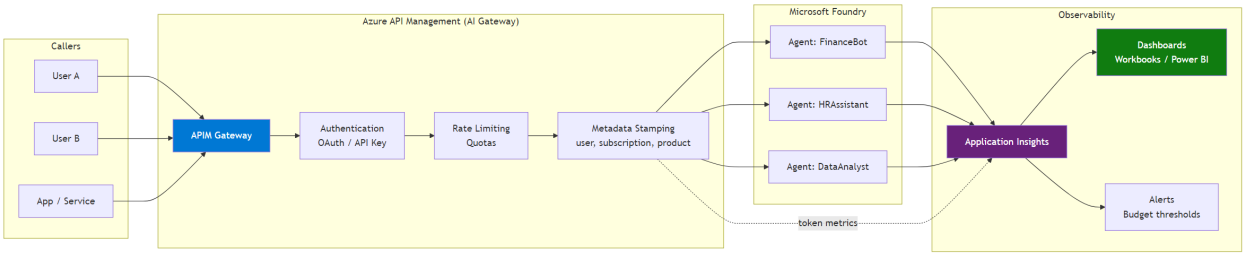

Figure 1: Diagram 1

1.3.1 Data Flow — Per-Request

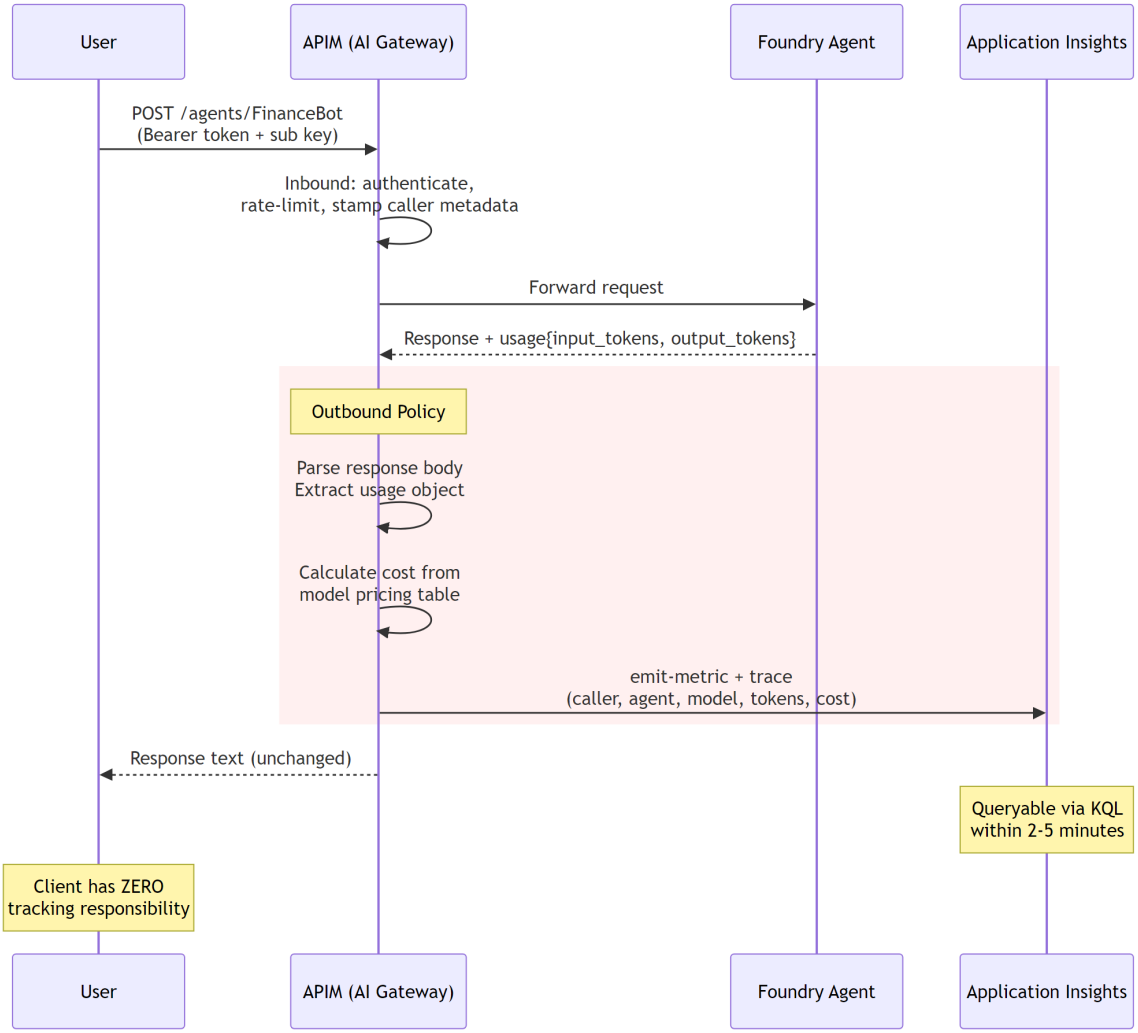

Figure 2: Diagram 2

1.4 Prerequisites

Component	Required
Azure AI Foundry project	With agents deployed (Prompt or Hosted)
Application Insights	Connected to the Foundry project (setup guide)
Azure API Management	Standard v2 or Premium (for `emit-metric` policy support)
APIM Diagnostic Settings	Enabled → sends `ApiManagementGatewayLogs` to App Insights
Python packages (optional, for querying)	`azure-identity`, `azure-monitor-query`, `httpx`
Azure CLI	Authenticated (`az login`)

```
# Client-side - only needed if you want to query usage programmatically
pip install httpx azure-identity azure-monitor-query python-dotenv
```

> **Important**: Foundry Agent Service does **not** impose rate limits on API calls. Rate limiting is applied only at the model deployment level (see limits). APIM adds the per-caller rate limiting layer described in Step 3.

> **Key difference from client-side approaches**: The client does NOT need `opentelemetry`, `azure-monitor-opentelemetry`, or any tracking SDK for per-caller attribution. Token extraction and metric emission happen entirely in APIM outbound policies. Client-side SDK tracing (Step 1) is for Foundry's built-in per-agent tracing — a complementary layer.

1.5 Step 1 — Connect Application Insights to Foundry

This step enables **Foundry's built-in tracing** (per-agent). APIM's per-caller tracking builds on top of this.

1. **Foundry portal** → your project → **Agents** → **Traces** tab
2. Click **Connect** → select or create an Application Insights resource
3. Alternatively: **Project details** → **Connected resources** → **Add connection** → **Application Insights**
4. Assign **Log Analytics Reader** role to users who need to query traces

   Once connected, Foundry automatically logs server-side traces for Prompt agents (GA), Hosted agents (preview), and Workflows (preview). Traces appear in the Foundry portal within 2–5 minutes and are retained for 90 days.

1.5.1 Optional: Client-Side Tracing (SDK)

For deeper visibility into your client application code, install the SDK tracing packages:

```
pip install azure-ai-projects azure-identity opentelemetry-sdk azure-core-tracing-opente
```

Then instrument your code:

```
import os
os.environ["AZURE_EXPERIMENTAL_ENABLE_GENAI_TRACING"] = "true"

from azure.ai.projects import AIProjectClient
from azure.ai.projects.telemetry import AIProjectInstrumentor
from azure.identity import DefaultAzureCredential
from azure.monitor.opentelemetry import configure_azure_monitor

# Get App Insights connection string from the project
client = AIProjectClient(endpoint=PROJECT_ENDPOINT, credential=DefaultAzureCredential())
conn_str = client.telemetry.get_application_insights_connection_string()
configure_azure_monitor(connection_string=conn_str)

# Enable GenAI tracing instrumentation
AIProjectInstrumentor().instrument()
```

> Client-side tracing captures spans for model calls, tool invocations, and custom logic. It supports trace context propagation (W3C) for distributed tracing across services.
>
> **Reference**: Client-side tracing · Framework integrations

---

1.6 Step 2 — Configure APIM as AI Gateway

1.6.1 2.1 — Create API routes to Foundry agents

For each agent, create an APIM operation that proxies to the Foundry Responses API:

```
Backend URL: https://<resource>.services.ai.azure.com/api/projects/<project>
```

1.6.2 2.2 — Configure authentication

APIM authenticates users via **subscription keys** or **OAuth 2.0**. Each subscription key maps to a user or team — this is the identity dimension in your telemetry.

1.6.3 2.3 — APIM Inbound Policy — Caller Metadata

Add this **inbound** policy to stamp every request with caller metadata:

```
<inbound>
    <base />
    <!-- Extract caller identity -->
    <set-variable name="caller-id"
        value="@(context.Subscription?.DisplayName ?? context.User?.Id ?? "anonymous")"
    <set-variable name="caller-product"
        value="@(context.Product?.DisplayName ?? "none")" />
```

```
    <!-- Forward to Foundry with managed identity -->
    <authentication-managed-identity resource="https://ai.azure.com" />

    <!-- Add trace headers -->
    <set-header name="X-Caller-Id" exists-action="override">
        <value>@((string)context.Variables["caller-id"])</value>
    </set-header>
    <set-header name="Ocp-Apim-Trace" exists-action="override">
        <value>true</value>
    </set-header>
</inbound>
```

1.6.4 2.4 — APIM Outbound Policy — Server-Side Token Extraction

This is the critical part. The **outbound** policy intercepts the Foundry response, parses the `usage` object, calculates cost, and emits metrics + traces to Application Insights — **without any client involvement**.

```
<outbound>
    <base />

    <!-- ═══ 1. Read and preserve the response body ═══ -->
    <set-variable name="responseBody"
        value="@(context.Response.Body.As<JObject>(preserveContent: true))" />

    <!-- ═══ 2. Extract token usage from Foundry response ═══ -->
    <set-variable name="input-tokens" value="@{
        var body = (JObject)context.Variables["responseBody"];
        return body["usage"]?["input_tokens"]?.ToString() ?? "0";
    }" />
    <set-variable name="output-tokens" value="@{
        var body = (JObject)context.Variables["responseBody"];
        return body["usage"]?["output_tokens"]?.ToString() ?? "0";
    }" />
    <set-variable name="total-tokens" value="@{
        var body = (JObject)context.Variables["responseBody"];
        return body["usage"]?["total_tokens"]?.ToString() ?? "0";
    }" />
    <set-variable name="model-name" value="@{
        var body = (JObject)context.Variables["responseBody"];
        return body["model"]?.ToString() ?? "unknown";
    }" />

    <!-- ═══ 3. Calculate estimated cost (USD) ═══ -->
    <set-variable name="cost-usd" value="@{
        var inTok  = int.Parse((string)context.Variables["input-tokens"]);
        var outTok = int.Parse((string)context.Variables["output-tokens"]);
```

```
        var model  = (string)context.Variables["model-name"];

        // Per-token pricing (USD) — update as pricing changes
        var inputPrice  = 0.002  / 1000.0;  // default
        var outputPrice = 0.008  / 1000.0;
        if (model.Contains("4o-mini"))  { inputPrice = 0.00015 / 1000.0; outputPrice = 0
        else if (model.Contains("4o")) { inputPrice = 0.0025  / 1000.0; outputPrice = 0.
        else if (model.Contains("5-mini")) { inputPrice = 0.0003 / 1000.0; outputPrice =

        var cost = (inTok * inputPrice) + (outTok * outputPrice);
        return cost.ToString("F6");
    }" />

    <!-- ═══ 4. Emit custom metrics to Azure Monitor ═══ -->
    <emit-metric name="FoundryAgent.TokenUsage"
                 value="@(int.Parse((string)context.Variables["total-tokens"]))"
                 namespace="FoundryAgents">
        <dimension name="CallerID"
                   value="@(context.Subscription?.DisplayName ?? context.User?.Id ?? "an
        <dimension name="AgentName" value="@(context.Api.Name)" />
        <dimension name="Model" value="@((string)context.Variables["model-name"])" />
    </emit-metric>

    <emit-metric name="FoundryAgent.CostUSD"
                 value="@(double.Parse((string)context.Variables["cost-usd"]))"
                 namespace="FoundryAgents">
        <dimension name="CallerID"
                   value="@(context.Subscription?.DisplayName ?? context.User?.Id ?? "an
        <dimension name="AgentName" value="@(context.Api.Name)" />
        <dimension name="Model" value="@((string)context.Variables["model-name"])" />
    </emit-metric>

    <!-- ═══ 5. Emit structured trace for per-request audit trail ═══ -->
    <trace source="foundry.agent.usage" severity="information">
        <message>@{
            return new JObject(
                new JProperty("caller_id",         context.Subscription?.DisplayName ??
                new JProperty("agent_name",        context.Api.Name),
                new JProperty("model",             (string)context.Variables["model-name
                new JProperty("input_tokens",      (string)context.Variables["input-toke
                new JProperty("output_tokens",     (string)context.Variables["output-tok
                new JProperty("total_tokens",      (string)context.Variables["total-toke
                new JProperty("cost_usd",          (string)context.Variables["cost-usd"]
                new JProperty("operation_id",      context.RequestId.ToString()),
                new JProperty("subscription_name", context.Subscription?.Name ?? "none")
                new JProperty("product_name",      context.Product?.DisplayName ?? "none
                new JProperty("client_ip",         context.Request.IpAddress)
```

```
            ).ToString();
        }</message>
    </trace>
</outbound>
```

Why this is better than client-side tracking:

Aspect	Client-side (old)	APIM Outbound (new)
Bypass risk	Any client that skips the SDK → no tracking	Every request through APIM is tracked
Client complexity	Must install OTel SDK + configure	Just call the API — zero instrumentation
Trust model	Client reports its own usage (honor system)	Gateway extracts from actual response
Language support	Python only (or rewrite per language)	Any HTTP client (curl, C#, JS, Power Automate)
Tamper-proof	Client can lie about token counts	APIM reads the real Foundry response

1.6.5 2.5 — APIM Subscription per user/team

Create subscriptions in APIM for each caller:

Subscription	Scope	Maps to
`sub-finance-team`	Product: Foundry Agents	Finance team
`sub-hr-team`	Product: Foundry Agents	HR team
`sub-app-crm`	Product: Foundry Agents	CRM application

Each subscription key becomes the `caller-id` in your telemetry.

1.7 Step 3 — Enable APIM Diagnostic Settings

For APIM to emit the `trace` and `emit-metric` data to Application Insights, you must configure diagnostic settings.

1.7.1 3.1 — Connect App Insights to APIM

1. **Azure Portal** → API Management → your APIM instance
2. **Application Insights** blade → **+ Add**
3. Select the same App Insights resource connected to Foundry
4. Set **Sampling** to `100%` for full audit trail (reduce in high-volume production)

1.7.2 3.2 — Enable API-level diagnostics

For each API (agent), enable Application Insights logging:

1. **APIs** → select your Foundry agent API
2. **Settings** → **Diagnostics** → select the App Insights logger
3. Enable **Request body** and **Response body** logging (optional, for debugging)

1.7.3 3.3 — Rate Limiting and Quotas (recommended)

Add rate limiting per subscription to prevent abuse:

```
<inbound>
    <!-- ... existing inbound policies ... -->

    <!-- Rate limit: max 100 calls per minute per subscription -->
    <rate-limit-by-key calls="100" renewal-period="60"
        counter-key="@(context.Subscription.Id)" />

    <!-- Quota: max 5000 calls and 50MB per day per subscription -->
    <quota-by-key calls="5000" bandwidth="50000" renewal-period="86400"
        counter-key="@(context.Subscription.Id)" />
</inbound>
```

1.7.4 3.4 — OAuth 2.0 Authentication (production)

For production deployments, replace subscription keys with **Entra ID OAuth 2.0**:

```
<inbound>
    <base />
    <!-- Validate JWT from Entra ID -->
    <validate-azure-ad-token tenant-id="{{tenant-id}}"
                             output-token-variable-name="jwt-token">
        <client-application-ids>
            <application-id>{{allowed-client-app-id}}</application-id>
        </client-application-ids>
    </validate-azure-ad-token>

    <!-- Extract caller from JWT claims -->
    <set-variable name="caller-id"
        value="@(((Jwt)context.Variables["jwt-token"]).Claims.GetValueOrDefault("preferr
    <set-variable name="caller-oid"
        value="@(((Jwt)context.Variables["jwt-token"]).Claims.GetValueOrDefault("oid", "

    <!-- Continue with managed identity to Foundry -->
    <authentication-managed-identity resource="https://ai.azure.com" />
</inbound>
```

> With OAuth, the `caller-id` dimension in your metrics is the actual Entra ID user principal name — no ambiguity.

1.8 Step 4 — KQL Queries for Monitoring

Run these in **Application Insights** → **Logs**. The data comes from APIM outbound `trace` policy — no client instrumentation needed.

1.8.1 Per-request audit trail

```
traces
| where message has "foundry.agent.usage"
| extend payload = parse_json(message)
| extend
    agent_name        = tostring(payload["agent_name"]),
    model             = tostring(payload["model"]),
    caller_id         = tostring(payload["caller_id"]),
    input_tokens      = toint(payload["input_tokens"]),
    output_tokens     = toint(payload["output_tokens"]),
    total_tokens      = toint(payload["total_tokens"]),
    cost_usd          = todouble(payload["cost_usd"]),
    client_ip         = tostring(payload["client_ip"]),
    subscription_name = tostring(payload["subscription_name"]),
    product_name      = tostring(payload["product_name"])
| project timestamp, caller_id, agent_name, model,
          input_tokens, output_tokens, total_tokens, cost_usd,
          client_ip, subscription_name, product_name
| order by timestamp desc
```

1.8.2 Token consumption by user (last 30 days)

```
traces
| where message has "foundry.agent.usage" and timestamp > ago(30d)
| extend payload = parse_json(message)
| summarize
    total_calls  = count(),
    total_tokens = sum(toint(payload["total_tokens"])),
    total_cost   = sum(todouble(payload["cost_usd"]))
  by caller_id = tostring(payload["caller_id"])
| order by total_cost desc
```

1.8.3 Token consumption by user × agent

```
traces
| where message has "foundry.agent.usage" and timestamp > ago(30d)
| extend payload = parse_json(message)
| summarize
```

```
    calls        = count(),
    total_tokens = sum(toint(payload["total_tokens"])),
    total_cost   = sum(todouble(payload["cost_usd"]))
  by caller_id   = tostring(payload["caller_id"]),
     agent_name  = tostring(payload["agent_name"]),
     model       = tostring(payload["model"])
| order by total_cost desc
```

1.8.4 Top 10 most expensive agents

```
traces
| where message has "foundry.agent.usage" and timestamp > ago(7d)
| extend payload = parse_json(message)
| summarize
    total_calls  = count(),
    avg_tokens   = avg(toint(payload["total_tokens"])),
    total_cost   = sum(todouble(payload["cost_usd"]))
  by agent_name = tostring(payload["agent_name"])
| top 10 by total_cost desc
```

1.8.5 Daily cost trend per agent

```
traces
| where message has "foundry.agent.usage" and timestamp > ago(30d)
| extend payload = parse_json(message)
| summarize
    daily_cost = sum(todouble(payload["cost_usd"]))
  by bin(timestamp, 1d),
     agent_name = tostring(payload["agent_name"])
| render timechart
```

1.8.6 Prompt vs Completion token ratio (optimization signal)

```
traces
| where message has "foundry.agent.usage" and timestamp > ago(7d)
| extend payload = parse_json(message)
| summarize
    avg_input      = avg(toint(payload["input_tokens"])),
    avg_output     = avg(toint(payload["output_tokens"])),
    ratio          = avg(todouble(payload["input_tokens"])) / avg(todouble(payload["outp
  by agent_name = tostring(payload["agent_name"])
| order by ratio desc
```

A high prompt/completion ratio may indicate verbose system prompts that could be optimized.

1.9 Step 5 — Dashboards and Alerts

1.9.1 Azure Workbook (recommended)

1. **Application Insights** → **Workbooks** → New
2. Add KQL query tiles for:
   - Token consumption by user (bar chart)
   - Daily cost trend (timechart)
   - Top agents by cost (table)
   - Caller × Agent matrix (heatmap)
3. Pin to Azure Dashboard for team visibility

1.9.2 Budget Alerts

Create an alert rule when a caller exceeds a token budget:

```
traces
| where message has "foundry.agent.usage" and timestamp > ago(1d)
| extend payload = parse_json(message)
| summarize daily_cost = sum(todouble(payload["cost_usd"]))
  by caller_id = tostring(payload["caller_id"])
| where daily_cost > 5.00
```

Configure: **Application Insights** → **Alerts** → **New Alert Rule** → Custom log search → threshold = 1 result.

1.9.3 Power BI Integration

Export telemetry to Power BI for executive reporting: 1. **Application Insights** → **Logs** → run your KQL query 2. Click **Export** → **Export to Power BI (M query)** 3. Open in Power BI Desktop → publish to workspace

---

1.10 Prompt Agent vs Hosted Agent — Compatibility

The APIM outbound policy approach works with **both** agent types, but with important nuances. This section explains the differences and edge cases.

1.10.1 What Are Hosted Agents in Foundry?

> **Important**: Hosted Agents are currently in **public preview**. Tracing for hosted agents is also in preview. Some features may change or have constraints. See limits and regions.

Hosted Agents are **not** raw containers you manage yourself. They are containerized agents that run on **Foundry Agent Service** with a managed hosting adapter (`azure-ai-agentserver-*`). The hosting adapter provides:

- **Automatic protocol translation** — wraps your agent code to expose the standard **Responses API** (same as Prompt Agents)
- **Built-in OpenTelemetry** — traces, metrics, and logs are exported automatically
- **Conversation orchestration** — message serialization, streaming, state management
- **Identity management** — runs with the project managed identity (or a dedicated agent identity when published)

The hosting adapter supports three frameworks:

Framework	Python	C#
Microsoft Agent Framework	Yes	Yes
LangGraph	Yes	No
Custom code	Yes	Yes

1.10.2 How Hosted Agents Make LLM Calls

In the standard pattern, hosted agents call models through the **Foundry project endpoint** — not directly to Azure OpenAI:

```
# Standard hosted agent code - uses Foundry project endpoint
from agent_framework import ChatAgent
from agent_framework.azure import AzureAIAgentClient

agent = ChatAgent(
    chat_client=AzureAIAgentClient(
        project_endpoint=PROJECT_ENDPOINT,         # Foundry project endpoint
        model_deployment_name="gpt-4.1",
        credential=DefaultAzureCredential(),
    ),
    instructions="You are a helpful assistant...",
    tools=[my_tool],
)
```

Since the LLM calls go through **Foundry's infrastructure** (not direct Azure OpenAI), and the hosting adapter returns responses in the standard **Responses API format**, the response includes the `usage` object — just like Prompt Agents.

1.10.3 Prompt Agent — Full Compatibility (GA)

Prompt Agents are fully orchestrated by Foundry. Tracing is **generally available (GA)**. The Responses API always returns `usage`:

```
{
  "id": "resp_abc123",
  "output": [{ "content": [{ "type": "output_text", "text": "..." }] }],
  "usage": {
    "input_tokens": 230,
    "output_tokens": 512,
    "total_tokens": 742
```

```
  },
  "model": "gpt-4.1"
}
```

The APIM outbound policy parses this response and emits metrics — **no changes needed**.

1.10.4 Hosted Agent (Standard Pattern) — Full Compatibility (Preview)

> **Note**: Tracing for hosted agents is in **public preview**. The Responses API format and `usage` object behavior may change.

When using the hosting adapter with `AzureAIAgentClient` (or any adapter-supported framework), the response is also returned in the **Responses API format**. Since the hosting adapter handles protocol translation, the `usage` object is included in the response.

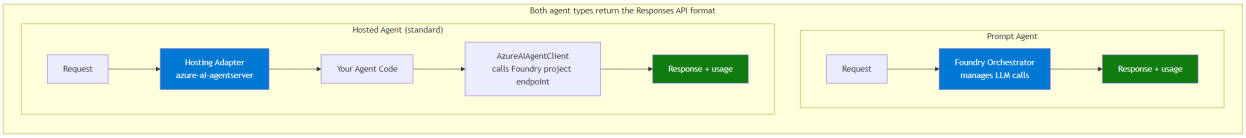

Figure 3: Diagram 3

Result: The same APIM outbound policy works for both Prompt and Hosted agents.

1.10.5 Edge Case: Hosted Agent with External LLM Calls

The only scenario where token tracking breaks is when a hosted agent **bypasses** the Foundry project endpoint and calls LLMs directly (e.g., direct Azure OpenAI, Anthropic API, or a self-hosted model):

```
# ⚠ Edge case - hosted agent bypasses Foundry
from openai import AzureOpenAI
client = AzureOpenAI(
    azure_endpoint="https://my-aoai.openai.azure.com",  # Direct call, bypasses Foundry
    api_key=os.environ["AOAI_KEY"],
)
```

In this case: 1. The hosting adapter cannot report `usage` for calls it doesn't manage 2. The Responses API response may have **partial or missing** `usage` data 3. APIM outbound policy has incomplete information

1.10.6 Solution for Edge Cases: Double-Hop APIM Pattern

If your hosted agent must call LLMs directly (not through the Foundry project endpoint), route those calls through APIM:

Hop 2 APIM Policy — use the built-in `azure-openai-emit-token-metric`:

```
<!-- API: Azure OpenAI (internal, called by hosted agents with direct LLM calls) -->
<inbound>
    <base />
    <set-variable name="caller-agent"
```

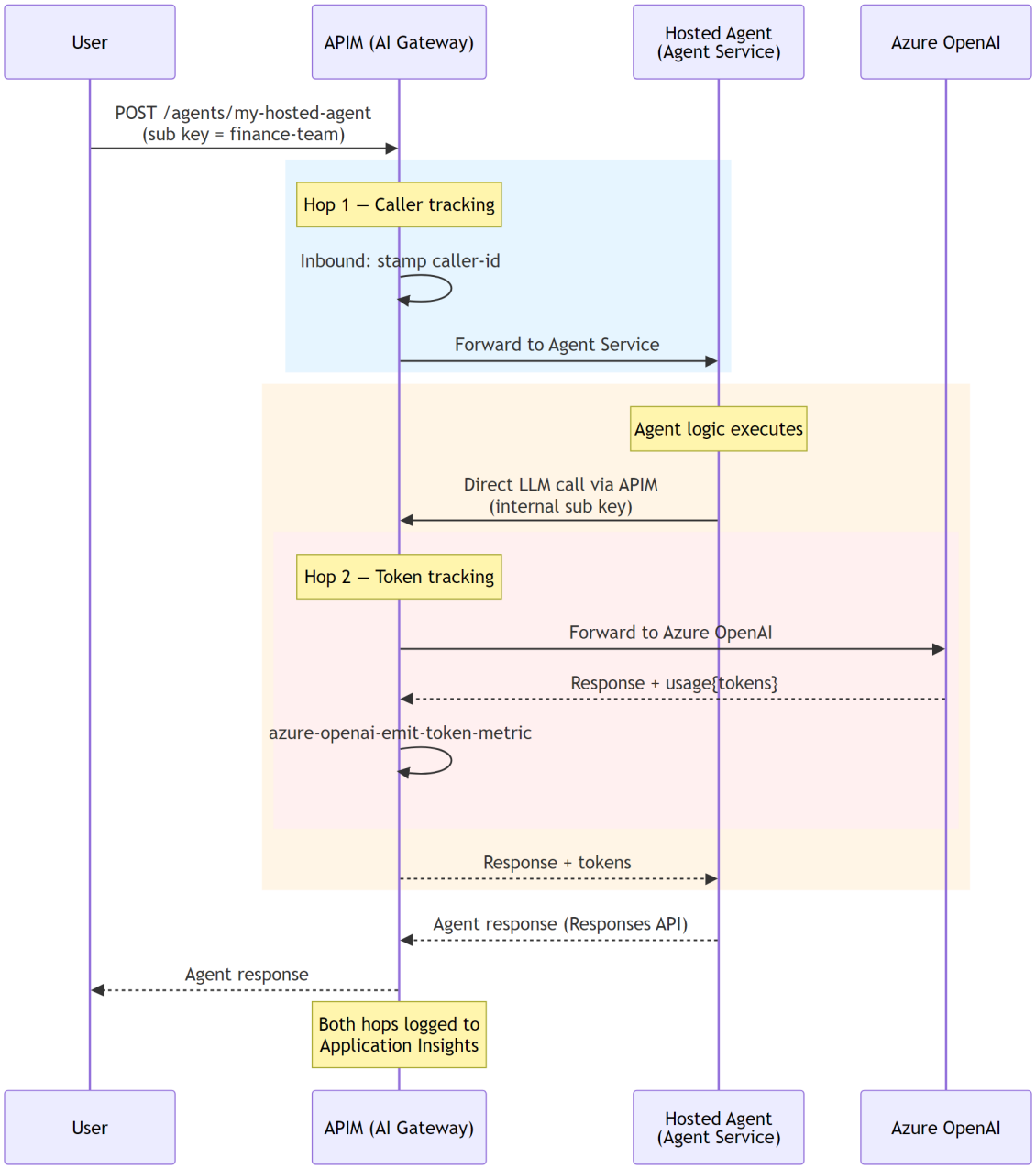

Figure 4: Diagram 4

```
        value="@(context.Request.Headers.GetValueOrDefault("X-Agent-Name", "unknown-agen
    <authentication-managed-identity resource="https://cognitiveservices.azure.com" />
</inbound>

<outbound>
    <base />
    <azure-openai-emit-token-metric namespace="FoundryAgents.LLM">
        <dimension name="AgentName"
                   value="@(context.Request.Headers.GetValueOrDefault("X-Agent-Name", "u
        <dimension name="CallerID"
                   value="@(context.Subscription?.DisplayName ?? "anonymous")" />
        <dimension name="Model"
                   value="@(context.Request.MatchedParameters["deployment-id"])" />
    </azure-openai-emit-token-metric>
</outbound>
```

Container configuration — point direct LLM calls to APIM:

```
# In your hosted agent code — redirect direct LLM calls through APIM
from openai import AzureOpenAI
client = AzureOpenAI(
    azure_endpoint=os.environ["APIM_BASE"],           # APIM, not direct AOAI
    api_key=os.environ["APIM_INTERNAL_SUB_KEY"],       # APIM subscription key
    default_headers={
        "X-Agent-Name": os.environ.get("AGENT_NAME", "my-hosted-agent"),
    },
)
```

Set these **environment variables** in your hosted agent definition:

```
agent = client.agents.create_version(
    agent_name="my-agent",
    definition=HostedAgentDefinition(
        # ... image, cpu, memory ...
        environment_variables={
            "AZURE_AI_PROJECT_ENDPOINT": PROJECT_ENDPOINT,
            "APIM_BASE": "https://my-apim.azure-api.net",
            "APIM_INTERNAL_SUB_KEY": "<subscription-key>",    # Use Key Vault connection
            "AGENT_NAME": "my-agent",
        }
    )
)
```

> **Security note**: Don't put the APIM subscription key directly in environment variables. Use a Key Vault connection as recommended by the Foundry documentation.

1.10.7 KQL: Correlating Both Hops (Edge Case Only)

When a hosted agent uses the double-hop pattern, correlate caller identity (Hop 1) with token usage (Hop 2):

```
// Hop 1: Caller -> Agent (from custom outbound trace)
let hop1 = traces
| where message has "foundry.agent.usage"
| extend p = parse_json(message)
| project timestamp, caller_id = tostring(p["caller_id"]),
          agent_name = tostring(p["agent_name"]),
          operation_id;

// Hop 2: Agent -> Azure OpenAI (from azure-openai-emit-token-metric)
let hop2 = customMetrics
| where name == "FoundryAgents.LLM.TokenUsage"
| extend agent = tostring(customDimensions["AgentName"]),
         model = tostring(customDimensions["Model"]),
         tokens = valueSum;

// Combine
hop2
| summarize total_tokens = sum(tokens) by agent, model
| join kind=inner (
    hop1 | summarize calls = count() by caller_id, agent_name
) on $left.agent == $right.agent_name
| project caller_id, agent_name, model, calls, total_tokens
| order by total_tokens desc
```

1.10.8 Summary: Compatibility Matrix

Scenario	Tracing Status	Responses API `usage`?	APIM Outbound Works?	Action Required
Prompt Agent	GA	Always present	Yes	None
Hosted Agent (standard — AzureAIAgentClient)	Preview	Present (hosting adapter)	Yes	None
Hosted Agent (LangGraph via adapter)	Preview	Present (hosting adapter)	Yes	None

Scenario	Tracing Status	Responses API `usage`?	APIM Outbound Works?	Action Required
Hosted Agent (direct Azure OpenAI calls)	Preview	May be missing	Partial	Route LLM calls through APIM (double-hop)
Hosted Agent (external LLM — Anthropic, etc.)	Preview	Missing	No	Route through APIM or implement custom logging

---

1.11 Full Python Solution

Since token tracking is handled entirely by APIM outbound policies, the client code is a **thin wrapper** — no OTel SDK, no tracking logic, just call the API.

1.11.1 Configuration (`.env`)

```
APIM_BASE=https://<your-apim>.azure-api.net
APIM_SUB_KEY=<your-subscription-key>
API_VERSION=2025-11-15-preview
APP_INSIGHTS_RESOURCE_ID=/subscriptions/<sub>/resourceGroups/<rg>/providers/Microsoft.In
```

1.11.2 `requirements.txt`

```
httpx
azure-identity
azure-monitor-query
python-dotenv
```

1.11.3 `agent_client.py` — Thin client (no tracking SDK)

```
"""
Foundry Agent Client - Call agents via APIM.
Token tracking is handled server-side by APIM outbound policies.

Usage:
    from agent_client import call_agent
    answer = call_agent("MyAgent", "Hello")
"""
```

```python
import os
import httpx
from dotenv import load_dotenv
from azure.identity import DefaultAzureCredential, get_bearer_token_provider

load_dotenv(override=True)

# ── Configuration ──────────────────────────────────────────────────────────
APIM_BASE    = os.environ["APIM_BASE"]
APIM_SUB_KEY = os.environ["APIM_SUB_KEY"]
API_VERSION  = os.getenv("API_VERSION", "2025-11-15-preview")

# ── Auth ───────────────────────────────────────────────────────────────────
credential = DefaultAzureCredential()
_token_fn = get_bearer_token_provider(credential, "https://ai.azure.com/.default")


def call_agent(
    agent_name: str,
    user_message: str,
    model: str = "gpt-4.1",
) -> str:
    """
    Call a Foundry agent via APIM.

    Token usage is automatically extracted and tracked by the APIM
    outbound policy — no client-side instrumentation needed.
    """
    token = _token_fn()

    r = httpx.post(
        f"{APIM_BASE}/foundry/models/{agent_name}",
        headers={
            "Authorization": f"Bearer {token}",
            "Ocp-Apim-Subscription-Key": APIM_SUB_KEY,
            "Content-Type": "application/json",
        },
        json={"model": model, "input": user_message},
        params={"api-version": API_VERSION},
        timeout=60,
    )
    r.raise_for_status()
    data = r.json()

    # Extract response text
    for item in data.get("output", []):
        for part in item.get("content", []):
```

```
            if part.get("type") == "output_text":
                return part["text"]

    return str(data)
```

1.11.4 Usage

```
# Just call the agent — APIM handles all tracking
answer = call_agent("FinanceBot", "Summarize Q4 earnings")
answer = call_agent("HRAssistant", "What is the PTO policy?")
answer = call_agent("FinanceBot", "Project revenue for Q1")

# The APIM subscription key identifies who you are.
# Token metrics are automatically emitted by the gateway.
# No OpenTelemetry. No tracking SDK. No client-side logic.
```

1.11.5 `query_usage.py` — Query token metrics from App Insights

```
"""Query token usage from Application Insights (emitted by APIM policies)."""

import os
from azure.identity import DefaultAzureCredential
from azure.monitor.query import LogsQueryClient
from dotenv import load_dotenv

load_dotenv(override=True)

APP_INSIGHTS_RESOURCE_ID = os.environ["APP_INSIGHTS_RESOURCE_ID"]

QUERIES = {
    "per_request": """
traces
| where message has "foundry.agent.usage"
| extend payload = parse_json(message)
| project
    timestamp,
    caller_id         = tostring(payload["caller_id"]),
    agent_name        = tostring(payload["agent_name"]),
    model             = tostring(payload["model"]),
    input_tokens      = toint(payload["input_tokens"]),
    output_tokens     = toint(payload["output_tokens"]),
    total_tokens      = toint(payload["total_tokens"]),
    cost_usd          = todouble(payload["cost_usd"]),
    client_ip         = tostring(payload["client_ip"])
| order by timestamp desc
""",
```

```
    "by_caller": """
traces
| where message has "foundry.agent.usage" and timestamp > ago(30d)
| extend payload = parse_json(message)
| summarize
    total_calls  = count(),
    total_tokens = sum(toint(payload["total_tokens"])),
    total_cost   = sum(todouble(payload["cost_usd"]))
  by caller_id = tostring(payload["caller_id"])
| order by total_cost desc
""",
    "by_caller_agent": """
traces
| where message has "foundry.agent.usage" and timestamp > ago(30d)
| extend payload = parse_json(message)
| summarize
    calls        = count(),
    total_tokens = sum(toint(payload["total_tokens"])),
    total_cost   = sum(todouble(payload["cost_usd"]))
  by caller_id  = tostring(payload["caller_id"]),
     agent_name = tostring(payload["agent_name"]),
     model      = tostring(payload["model"])
| order by total_cost desc
""",
}


def run_query(query_name: str = "per_request"):
    client = LogsQueryClient(DefaultAzureCredential())
    resp = client.query_resource(
        resource_id=APP_INSIGHTS_RESOURCE_ID,
        query=QUERIES[query_name],
        timespan=None,
    )
    if resp.status != "Success":
        raise RuntimeError(f"Query failed: {resp.status}")

    table = resp.tables[0]
    rows = [dict(zip(table.columns, r)) for r in table.rows]
    return rows


if __name__ == "__main__":
    import sys

    query = sys.argv[1] if len(sys.argv) > 1 else "per_request"
    rows = run_query(query)
```

```
    if not rows:
        print("No telemetry found.")
    else:
        print(f"Found {len(rows)} records\n")
        for r in rows[:20]:
            print(r)
```

1.12 Enterprise Considerations

1.12.1 Capability Hosts

For production deployments with compliance or data sovereignty requirements, configure **capability hosts** to control where agent data is stored:

Resource	Purpose	Azure Service
Thread storage	Conversation history, agent definitions	Azure Cosmos DB
File storage	Uploaded documents, blob storage	Azure Storage Account
Vector store	Embeddings and retrieval	Azure AI Search

Capability hosts are configured at account and project levels via REST API. Without them, Agent Service uses Microsoft-managed resources.

> **Reference**: Capability hosts

1.12.2 Framework-Specific Tracing

If your agents use specific frameworks, Foundry provides native tracing integrations that complement APIM's per-caller tracking:

Framework	Tracing Support	Additional Setup
Microsoft Agent Framework	Automatic — no code changes	None
LangChain / LangGraph	Via `langchain-azure-ai` package	`AzureAIOpenTelemetryTracer` callback
OpenAI Agents SDK	Via `AIProjectInstrumentor`	Set `AZURE_EXPERIMENTAL_ENABLE_GENAI_TRACING=`
Custom code	Manual with `@trace_function` decorator	Add decorator to key functions

These integrations emit OpenTelemetry-compliant spans (using GenAI semantic conventions) that appear in the Foundry portal **Traces** view alongside APIM's caller-attributed metrics.

Reference: Tracing integrations

1.12.3 Security and Privacy

Tracing can capture sensitive information. Follow these practices:

- **Don't store secrets** in prompts, tool arguments, or span attributes
- **Redact PII** before it enters telemetry — enable content recording (`OTEL_INSTRUMENTATION_GENAI_CAPTUR` only in development
- **Configure retention policies** on Application Insights to comply with GDPR / data retention requirements
- **Treat trace data as production telemetry** — apply the same access controls you use for logs and metrics

1.12.4 Observability Layers Summary

Layer	What It Tracks	Per-Agent	Per-Caller	Setup
Foundry built-in tracing	Inputs, outputs, tools, tokens, latency	Yes	No	Connect App Insights to project
Client-side SDK tracing	Model calls, tool invocations, custom spans	Yes	No	`AIProjectInstrumentor`
APIM outbound policy (this guide)	Tokens + cost correlated with caller identity	Yes	**Yes**	APIM as AI Gateway + outbound policies
APIM diagnostics	Request/response logs, HTTP status, latency	Yes	**Yes**	APIM → App Insights diagnostic settings

---

1.13 References

Resource	Link
Trace Agent Concepts	learn.microsoft.com
Set Up Tracing for Agents	learn.microsoft.com
Client-Side Tracing (Python/C#)	learn.microsoft.com
Framework Tracing Integrations	learn.microsoft.com
Agent Limits, Quotas, and Regions	learn.microsoft.com
Capability Hosts	learn.microsoft.com
Hosted Agents	learn.microsoft.com

Resource	Link
Agent Monitoring Dashboard	learn.microsoft.com
Tracking Every Token — Tech Community	techcommunity.microsoft.com
Full sample repo (APIM + App Insights)	github.com/ccoellomsft

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